# Edexcel A Level Physics

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## Black Body Radiation

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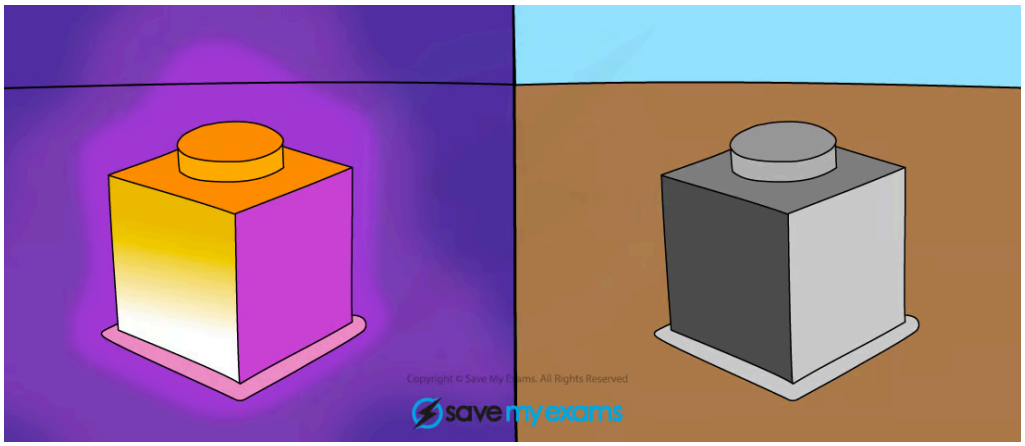
# Black Body Radiation

- Black body radiation is the name given to the **thermal radiation** emitted by all bodies (objects)
- All objects, no matter what temperature, emit black body radiation in the form of electromagnetic waves
- These electromagnetic waves usually lie in the **infrared** region of the spectrum but could be emitted in the form of visible light or other wavelengths, depending on the temperature
- The **hotter** object, the **more** infrared radiation it radiates in a given time




***The infrared emitted from a hot object can be detected using a special camera***

- A perfect black body is defined as:

  **An object that absorbs (or emits) all of the radiation incident on it and does not reflect or transmit any radiation**

- Since a good absorber is also a good emitter, a perfect black body would be the best possible emitter too
- As a result, an object which perfectly absorbs all radiation will be black
  - This is because the colour black is what is seen when **all** colours from the visible light spectrum are absorbed

## Absorption and Emission For Different Colours Table

| Colour | Absorbing | Emitting |
|---|---|---|
| Black | Good absorber | Good emitter |
| Dull/Dark | Reasonable absorber | Reasonable emitter |
| White | Poor absorber | Poor emitter |
| Shiny | Very poor absorber (Therefore a good reflector) | Very poor emitter |

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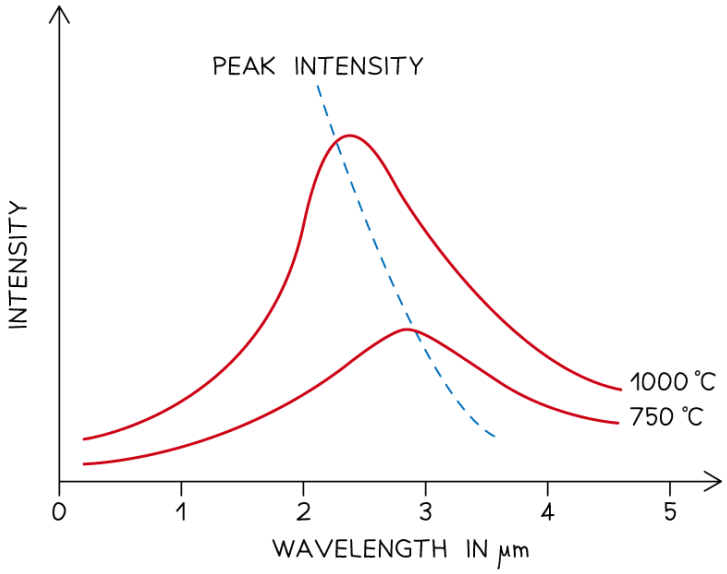
# Black Body Radiation Curves

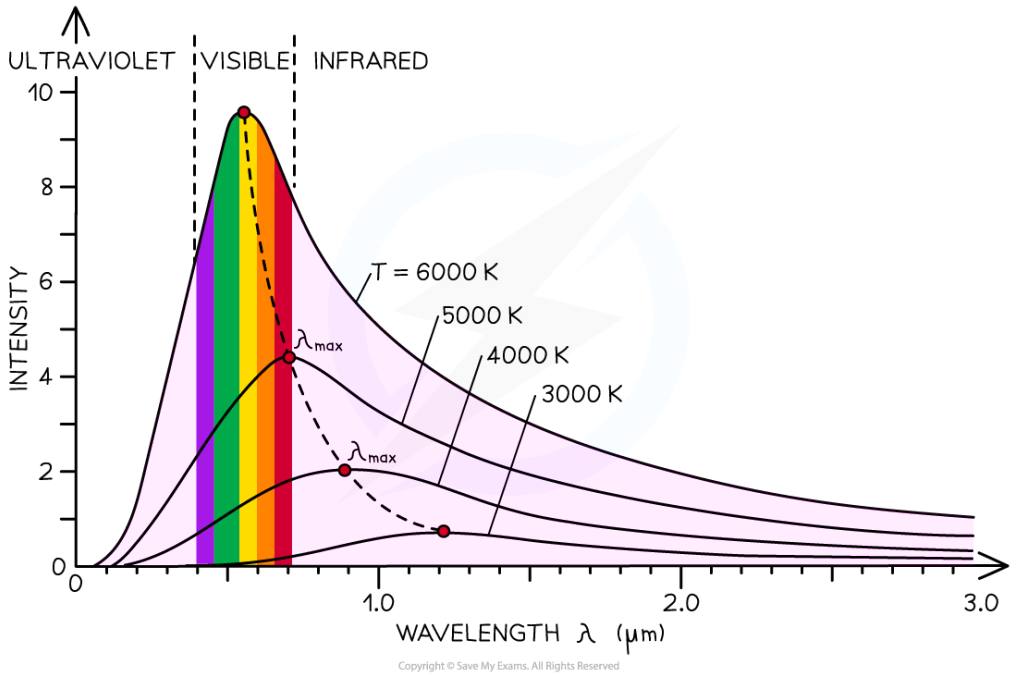
- All bodies (objects) emit a spectrum of thermal radiation in the form of electromagnetic waves
- The **intensity** and **wavelength** distribution of any emitted waves depends on the **temperature** of the body
- This is represented on a black body radiation curve
  - As the temperature increases, the peak of the curve moves




***Black body spectrum for objects of different temperatures***

- From the electromagnetic spectrum, waves with a **smaller** wavelength have **higher** energy (e.g. UV rays, X-rays)
- When an object gets hotter, the amount of **thermal radiation** it emits increases

- This increases the thermal energy emitted and therefore the **wavelength** of the emitted radiation **decreases**
- An ideal black body radiator is one that **absorbs and emits all wavelengths**
  - This idealised black body is a theoretical object
  - However, **stars** are the best approximation there is
- The radiation emitted from a black body has a **characteristic spectrum** that is determined by the temperature alone




***The intensity-wavelength graph shows how thermodynamic temperature links to the peak wavelength for four different bodies***

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# Stefan-Boltzmann Law

- An objects luminosity depends on two factors:
  - Its surface temperature
  - Its surface area
- The relationship between these is known as the **Stefan-Boltzmann Law**, which states:

  **The total energy emitted by a black body per unit area per second is proportional to the fourth power of the absolute temperature of the body**
- So, $\mathbf{L \propto AT^4}$
- The Stefan-Boltzmann Law equation is:

$$L = \sigma A T^4$$

- Where:
  - $L$ = luminosity of the star (W)
  - $A$ = surface area of the star
  - $\sigma$ = the Stefan-Boltzmann constant
  - $T$ = surface temperature of the star (K)
- The surface area of a star (or other spherical object) can be calculated using:

$$A = 4\pi r^2$$

- Where:
  - $r$ = radius of the star

### Worked Example

A camel has a body temperature of 40°C and a surface area of 16 $m^2$. The peak wavelength of the emitted spectrum from the camel is $\lambda_{max} = 8.6 \times 10^{-6}$ m. Calculate the total power radiated by the camel.

**Answer:**

**Step 1: List the known quantities**

- Surface area, $A = 16\ m^2$
- Temperature (in K), $T = 40 + 273 = 313$ K
- Stefan-Boltzmann constant, $\sigma = 5.67 \times 10^{-8}\ W\ m^{-2}\ K^{-4}$

**Step 2: Write down the Stefan-Boltzmann equation**


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$$L = \sigma A T^4$$

**Step 3: Substitute in the values**

$$L = (5.67 \times 10^{-8}) \times 16 \times 313^4 = 8707 \text{ W}$$

**Step 4: Write the answer to correct significant figures and include units**

Luminosity (power emitted) of the camel = 8700 W (2 s.f.)

## Examiner Tips and Tricks

Remember to convert temperatures into Kelvin.

If you are given the radius of a spherical object then its surface area *A* can be calculated using $\mathbf{A = 4\pi r^2}$ for the radius of the object *r*.

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# Wien's Law

- Wien's displacement law relates the observed wavelength of light from an object to its surface temperature, it states:

  **The black body radiation curve for different temperatures peaks at a wavelength that is inversely proportional to the temperature**

- This relation can be written as:

$$\lambda_{max} \propto \frac{1}{T}$$

- $\lambda_{max}$ is the maximum wavelength (m) emitted by an object at the peak intensity
- $T$ is the surface temperature (K) of an object
- Wien's Law equation is given by:

$$\lambda_{max} T = 2.9 \times 10^{-3} \text{ m K}$$

- This equation tells us the higher the temperature of a body:
  - The shorter the wavelength at the peak intensity, so **hotter** objects tend to be **white or blue**, and **cooler** objects tend to be **red or yellow**
  - The greater the intensity of the radiation at each wavelength

## Table to compare surface temperature and star colour

| Colour of star | Temperature / K |
|---|---|
| blue | > 33 000 |
| blue–white | 10 000 – 30 000 |
| white | 7 500 – 10 000 |
| yellow–white | 6 000 – 7 500 |
| yellow | 5 000 – 6 000 |
| orange | 3 500 – 5 000 |
| red | < 3 500 |

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## Worked Example

The spectrum of the star Rigel in the constellation of Orion peaks at a wavelength of 263 nm, while the spectrum of the star Betelgeuse peaks at a wavelength of 828 nm. Which of these two stars is cooler, Betelgeuse or Rigel?

**Answer:**

**Step 1: Write down Wien's displacement law**

$$\lambda_{max}T = 2.9 \times 10^{-3} \text{ m K}$$

**Step 2: Rearrange for temperature T**

$$T = \frac{2.9 \times 10^{-3}}{\lambda_{max}}$$

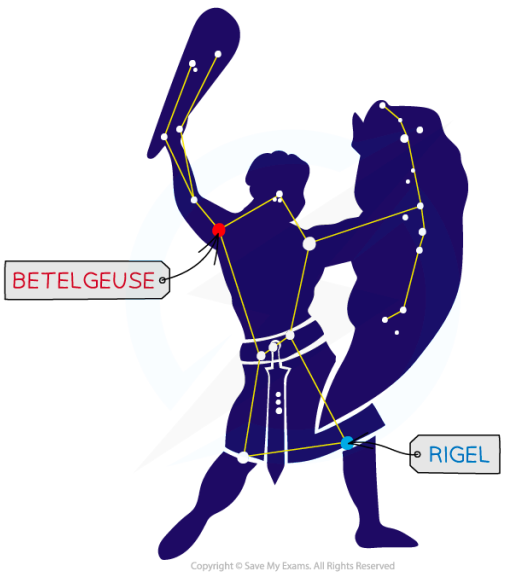
**Step 3: Calculate the surface temperature of each star**

$$\text{Rigel:} \quad T = \frac{2.9 \times 10^{-3}}{\lambda_{max}} = \frac{2.9 \times 10^{-3}}{263 \times 10^{-9}} = 11026 = \mathbf{11\ 000\ K}$$

$$\text{Betelguese:} \quad T = \frac{2.9 \times 10^{-3}}{\lambda_{max}} = \frac{2.9 \times 10^{-3}}{828 \times 10^{-9}} = 3502 = \mathbf{3\ 500\ K}$$

**Step 4: Write a concluding sentence**

Betelgeuse has a surface temperature of 3500 K, therefore, it is much cooler than Rigel

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*The Orion Constellation; cooler stars, such as Betelguese, appear red or yellow, while hotter stars, such as Rigel, appear white or blue*

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### Examiner Tips and Tricks

Note that the temperature used in Wien's Law is in **Kelvin** (K). Remember to convert from °C if the temperature is given in degrees in the question before using the Wien's Law equation.